Sistemas Distribuidos: Arquitecturas

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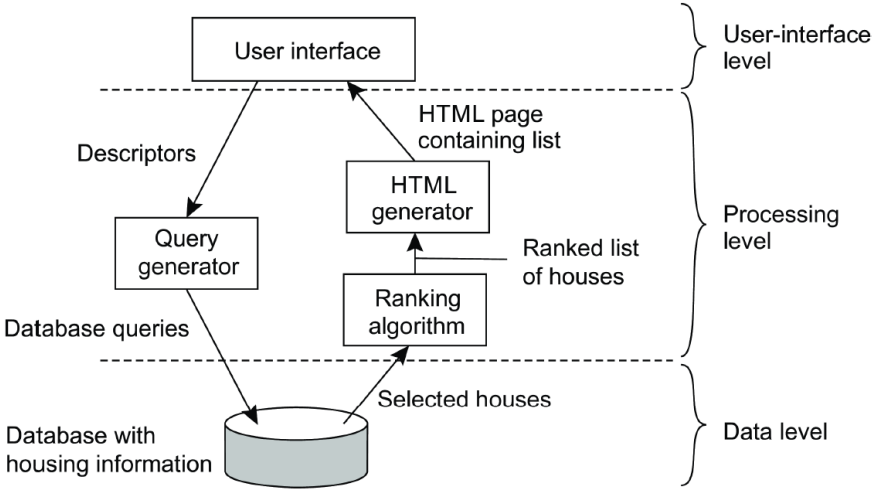
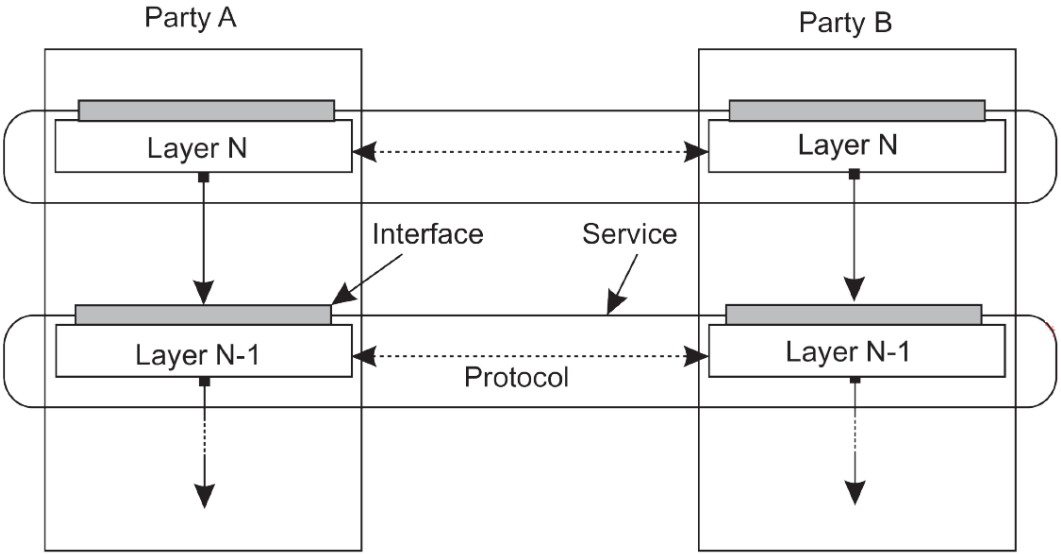
Clasificaciones generales

- Arquitecturas por capas
- Arquitecturas orientadas a servicios
- Arquitecturas Pub-Sub

Arquitecturas por capas

- Fuerte dependencia entre componentes
- Llamados desde las capas superiores a las inferiores
- En algunas ocasiones se realizan llamados desde capas inferiores a superiores

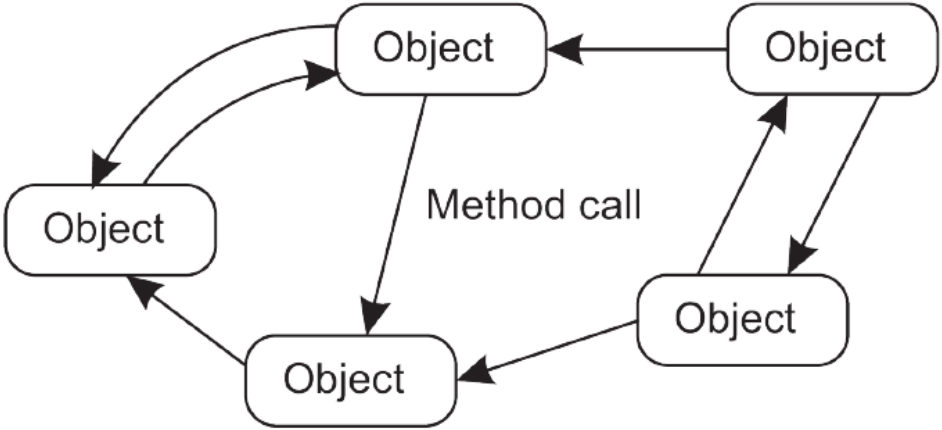

User interface
User-interface level
HTML page containing list
Descriptors
HTML generator
Query generator
Ranked list of houses
Processing level
Ranking algorithm
Database queries
Selected houses
Database with housing information
Data level

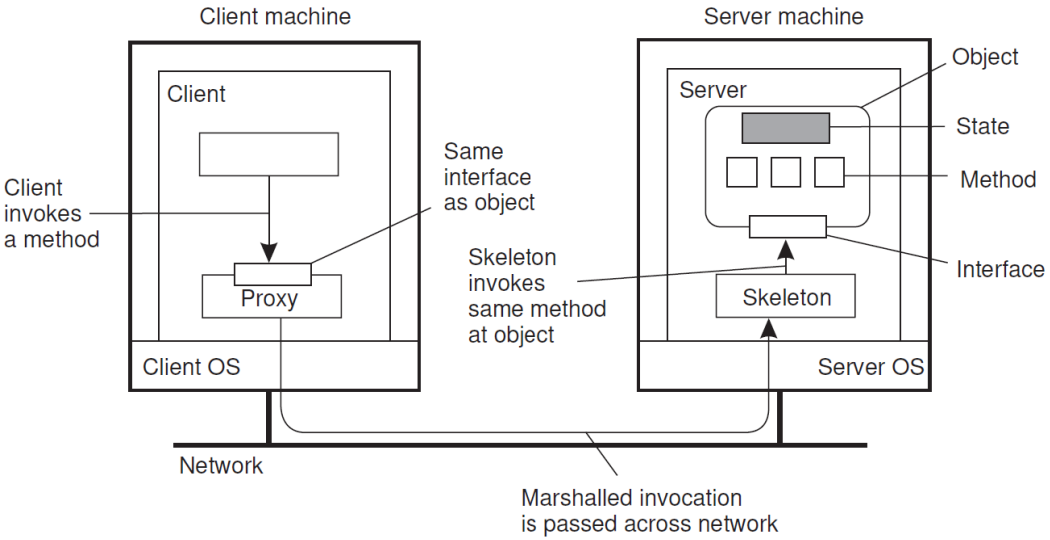

Party A
Party B
Layer N
Layer N
Interface
Service
Layer N-1
Layer N-1
Protocol

Arquitectura orientada a servicios

- Modularidad
- Cada componente puede ser reemplazada de forma sencilla
- Cada servicio se ejecuta en un hilo/proceso/sistema separado

Objetos remotos




Client machine
Server machine
Client
Server
Object
State
Method
Interface
Client invokes a method
Same interface as object
Skeleton invokes same method at object
Proxy
Skeleton
Client OS
Server OS
Network
Marshalled invocation is passed across network

Representational State Transfer (REST)

- Un solo esquema de nombrado
- Todos los servicios ofrecen la misma interfaz
- Los mensajes son autodescriptivos
- Luego de la ejecución de una operación, el componente olvida toda información sobre quién lo invocó

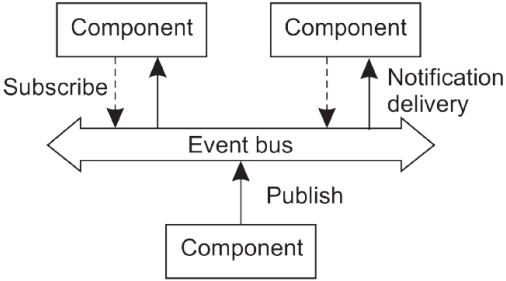
Operation	Description
PUT	Modify a resource by transferring a new state
POST	Create a new resource
GET	Retrieve the state of a resource in some representation
DELETE	Delete a resource

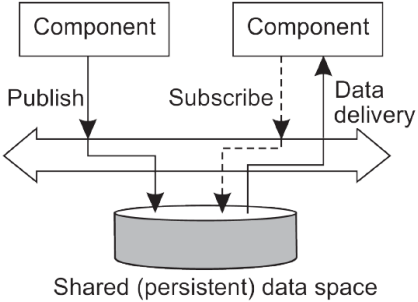
Arquitecturas Publisher-Subscriber

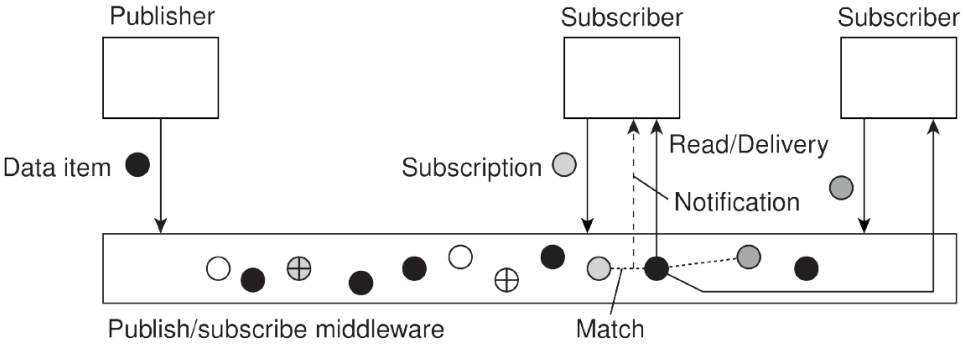
- Mínima dependencia entre componentes
- Independencia temporal de la comunicación de cada componente
- Se comparte un medio común para el intercambio de mensajes

Coordinación

	Temporally coupled	**Temporally decoupled**
Referentially coupled	Direct	Mailbox
Referentially decoupled	Event-based	Shared data space

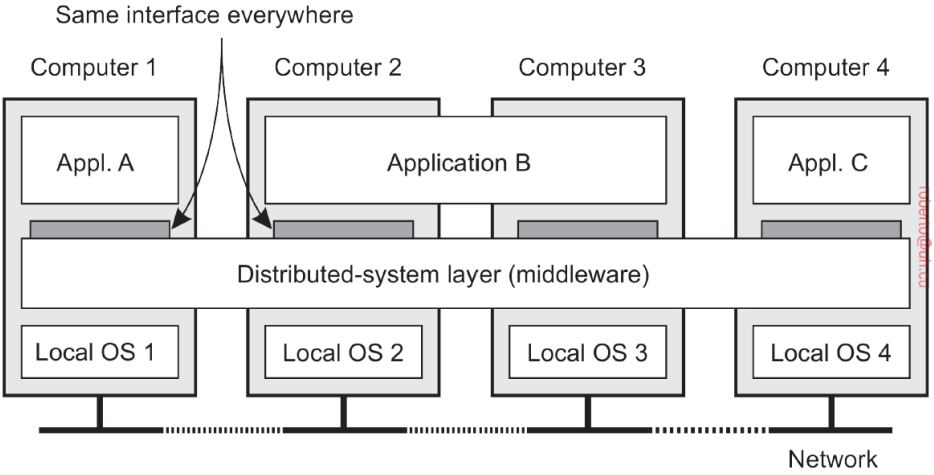

Component
Component
Subscribe
Notification
delivery
Event bus
Publish
Component


Component
Component
Publish
Subscribe
Data
delivery
Shared (persistent) data space


Publisher
Subscriber
Subscriber
Data item
Subscription
Read/Delivery
Notification
Publish/subscribe middleware
Match

Middleware

- Capa de software que abstrae componentes de un sistema
- Enmascara y se recupera de fallas
- Facilidades de comunicación y acceso a datos
- Brinda servicios de seguridad



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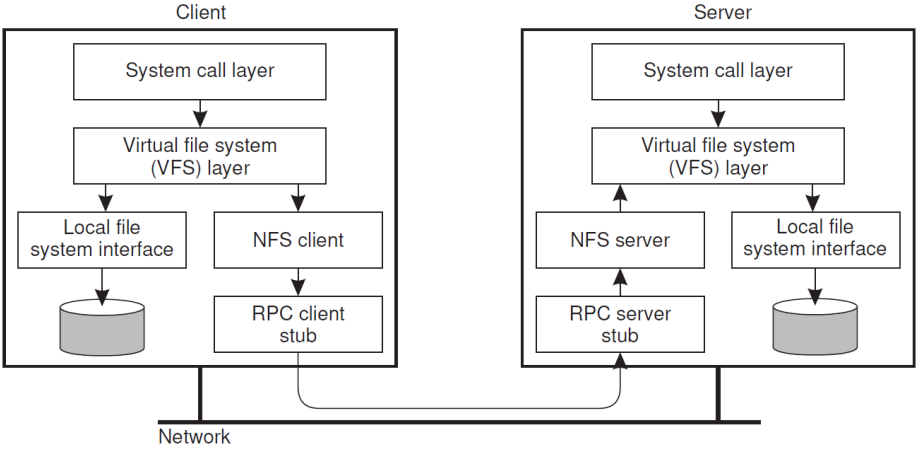
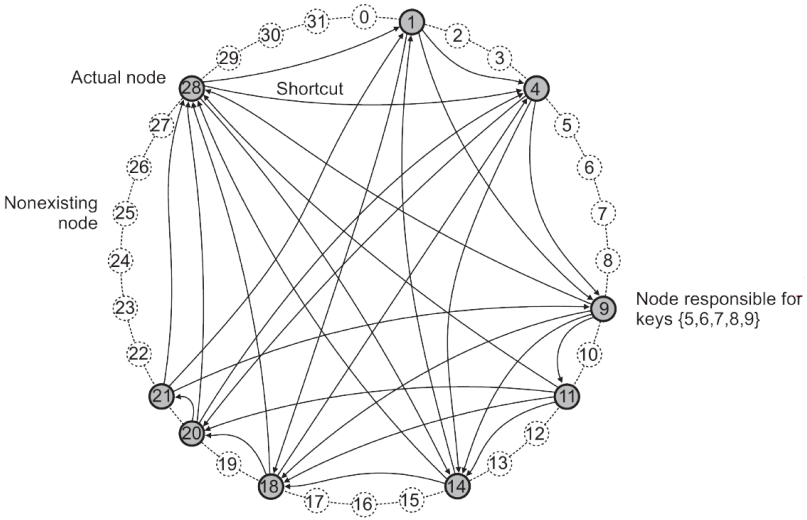
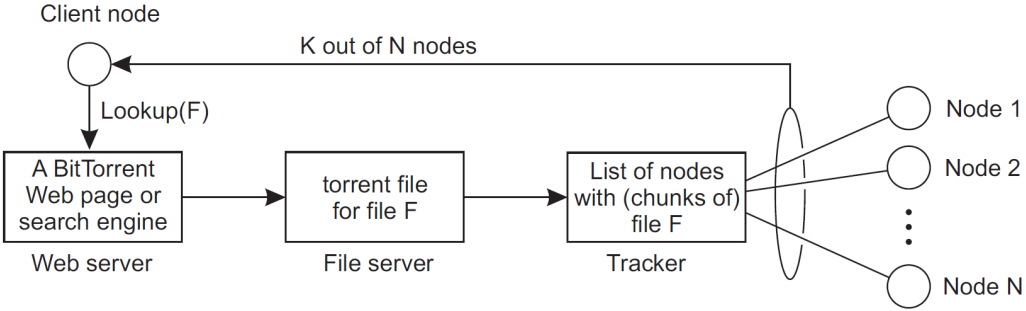
Ejemplos. Capas. NFS



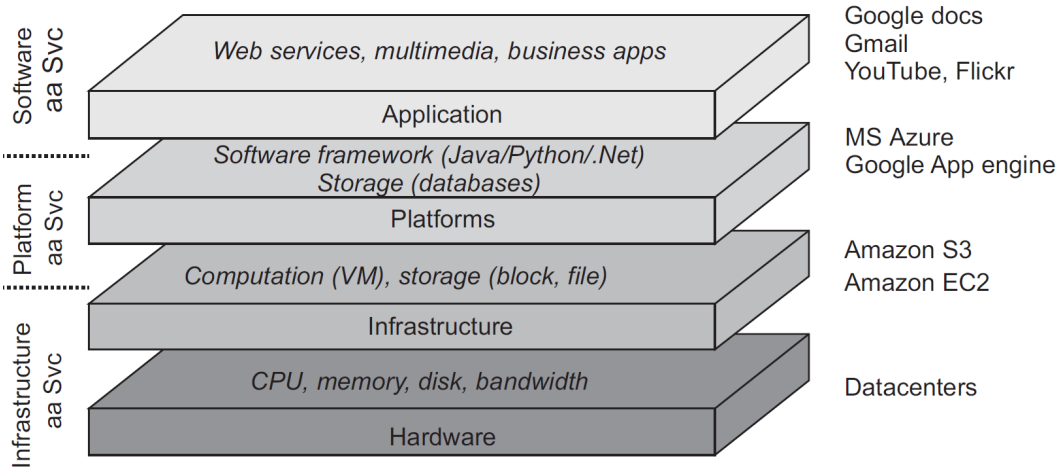
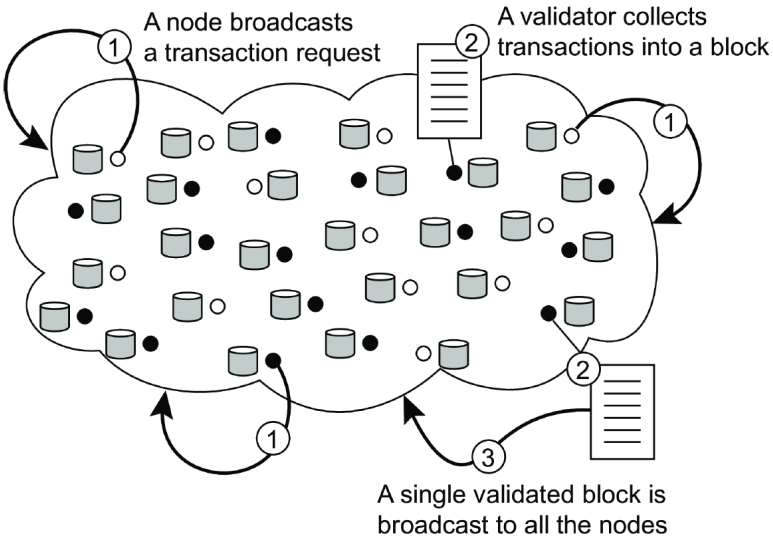
Ejemplos. Peer to peer estructurado. Chord



Ejemplos. Peer to peer sin esctructura. Bit-Torrent



Ejemplos. Arquitecturas híbridas. Cloud



Ejemplos. Arquitecturas híbridas. Blockchain